

## MODULE 01 / 40

# Northern Cardinal

*Cardinalis cardinalis***QUICK STATS**

|          |                                 |
|----------|---------------------------------|
| Length   | 8.3 – 9.3 in (21 – 23.5 cm)     |
| Wingspan | 9.8 – 12.2 in (25 – 31 cm)      |
| Weight   | 1.5 – 1.7 oz (42 – 48 g)        |
| Lifespan | Up to 15 years (avg. 3 in wild) |

**PRESENTATION**

The Northern Cardinal is one of the most iconic and instantly recognizable birds visiting North American gardens year-round. The brilliant scarlet male, with his dramatic crest and orange-red bill, is among the most photographed birds on the continent. The female, dressed in warm buff-brown with reddish tinges on her crest, wings and tail, is no less elegant. Both sexes remain on their territory throughout winter, bringing color and song to snow-covered gardens when most other species have migrated south.

**HABITAT & RANGE**

The Northern Cardinal thrives across the eastern half of North America, from southern Maine and the Great Lakes south through Florida, and westward to the Great Plains of Texas and Oklahoma. Populations are also well-established in Arizona, New Mexico, and parts of California. Southern Ontario and Quebec host significant breeding populations. The species has expanded its range northward over the past century, likely aided by the proliferation of bird feeders and milder winters. It favors woodland edges, dense shrubby thickets, hedgerows, suburban gardens, and parks with mature trees and dense understory cover.

**DIET & FEEDING**

- Black-oil sunflower seeds — the undisputed favorite
- Safflower seeds — especially attractive to cardinals
- White millet and cracked corn
- Wild berries: dogwood, holly, elderberry, sumac
- Insects and larvae (essential protein during breeding season)
- Fruit: grapes, mulberries, raspberries

BEHAVIOR & SOCIAL LIFE

Cardinals are territorial songbirds that maintain year-round home ranges of roughly 2–4 acres. Males fiercely defend their territory through song and, famously, by attacking their own reflection in windows — mistaking it for a rival. They are monogamous and pairs often remain together through multiple breeding seasons. Outside the breeding season, cardinals may gather in loose flocks of up to 70 individuals at abundant food sources. Males frequently feed females during courtship — a behavior called mate-feeding — passing seeds beak-to-beak.

NESTING & BREEDING

Breeding season runs from March through September, with females typically raising 2 to 4 broods per year — one of the highest rates among North American songbirds. The female constructs a cup-shaped nest over 3–9 days using twigs, bark strips, grasses and leaves, lined with fine grasses and hair. Nests are built 1–15 feet above ground in dense shrubs, vines or low trees. Each clutch contains 2–5 eggs (usually 3), pale whitish or greenish with brown speckles. Incubation lasts 11–13 days, performed almost exclusively by the female. Nestlings fledge at 9–11 days and the male takes over their care while the female begins a new nest.

SONG & COMMUNICATION

The Northern Cardinal is renowned for its rich, loud whistled song — a series of clear slurred notes often described as "cheer-cheer-cheer" or "birdy-birdy-birdy." Both males and females sing, which is unusual among North American songbirds. The female often sings from the nest, possibly communicating food needs to her mate. Cardinals have a repertoire of 16 or more distinct song patterns. Their sharp metallic "chip" call is an alarm note used to warn of predators. Song peaks in late winter and early spring as males establish territories.

CONSERVATION STATUS

|                |               |                   |                     |
|----------------|---------------|-------------------|---------------------|
| IUCN Red List: | Least Concern | Population trend: | Stable / Increasing |
|----------------|---------------|-------------------|---------------------|

With an estimated population of over 100 million individuals, the Northern Cardinal is one of the most abundant songbirds in North America. Its range has expanded northward significantly since the 1900s. It is the official state bird of 7 US states — more than any other species.

FEEDER TIPS

- Best feeder type: hopper feeder or large platform feeder
- Preferred seeds: safflower (squirrels tend to avoid it) and black-oil sunflower
- Cardinals feed most actively at dawn and dusk
- Place feeders near dense shrubs — cardinals prefer cover close by
- Offer a shallow birdbath: cardinals bathe and drink regularly

COLORING GUIDE

| ZONE            | RECOMMENDED COLOR       |
|-----------------|-------------------------|
| Male body       | Vivid red — cadmium red |
| Facial mask     | Deep black              |
| Bill            | Orange-coral            |
| Female body     | Warm sandy brown        |
| Wings & crest ■ | Rusty-red highlights    |
| Eye             | Dark brown, black ring  |

# COLOR ME!

Northern Cardinal • *Cardinalis cardinalis* • Module 01



## MODULE 02 / 40

# Black-capped Chickadee

*Poecile atricapillus***QUICK STATS**

|          |                                   |
|----------|-----------------------------------|
| Length   | 4.7 – 5.9 in (12 – 15 cm)         |
| Wingspan | 6.3 – 8.3 in (16 – 21 cm)         |
| Weight   | 0.3 – 0.5 oz (9 – 14 g)           |
| Lifespan | Up to 12 years (avg. 2–3 in wild) |

**PRESENTATION**

The Black-capped Chickadee is one of the most beloved and recognizable backyard birds across northern North America. Bold, curious, and remarkably tolerant of humans, it is often the first species to discover a newly installed feeder. Its cheerful "chick-a-dee-dee-dee" call is one of the most familiar sounds of North American winters. Despite its tiny size, the chickadee is supremely adapted to cold climates, capable of surviving temperatures well below freezing through a combination of extraordinary memory, metabolic flexibility, and dense winter plumage.

**HABITAT & RANGE**

The Black-capped Chickadee occupies a vast range across the northern tier of North America, from Alaska and the Yukon east to Newfoundland and Nova Scotia, and south through the northern United States including New England, the Great Lakes region, and the Rocky Mountain states down to northern New Mexico. It inhabits deciduous and mixed forests, forest edges, riparian corridors, suburban parks, and wooded gardens. It is a permanent resident throughout its range and does not migrate, relying instead on cached food and physiological adaptations to endure harsh winters.

**DIET & FEEDING**

- Black-oil sunflower seeds — the top choice at any feeder
- Suet cakes — critical high-energy winter food source
- Nyjer (thistle) seed and hulled sunflower chips
- Peanut pieces and peanut butter mixtures
- Mealworms (live or dried) during breeding season
- Wild: insects, insect eggs, spiders, berries, seeds

BEHAVIOR & SOCIAL LIFE

Chickadees are highly social birds that form stable winter flocks of 6–12 individuals with a clear dominance hierarchy. These flocks often serve as the nucleus for mixed-species foraging groups, attracting nuthatches, woodpeckers, and warblers. Within the flock, dominant birds feed first and claim the best roost sites. The chickadee's "chick-a-dee" alarm call is remarkably sophisticated: the number of "dee" notes appended to the call signals the level of threat, with more notes indicating a more dangerous predator nearby. Researchers have identified it as one of the most complex vocalizations in the animal kingdom.

NESTING & BREEDING

Breeding pairs form in late winter as flocks begin to break up. Chickadees are cavity nesters, excavating their own nest holes in soft rotting wood or using natural cavities and nest boxes. The female lines the cavity with moss, plant fibers, fur, and feathers, creating a deep insulating cup. Clutches typically contain 6–8 white eggs with reddish-brown speckles. Incubation lasts 12–13 days, performed by the female alone while the male brings her food. Both parents feed the nestlings, which fledge at 16 days. Pairs raise one brood per year.

SONG & COMMUNICATION

The chickadee has two primary vocalizations. Its contact and alarm call — the iconic "chick-a-dee-dee-dee" — varies in complexity based on context and threat level. Its territorial song is a clear, whistled two-note "fee-bee" (or "fee-bee-bee"), the first note higher than the second, heard most often on mild late-winter days as males begin to establish breeding territories. Chickadees also produce a variety of soft gargling calls within their flocks.

CONSERVATION STATUS

|                |               |                   |        |
|----------------|---------------|-------------------|--------|
| IUCN Red List: | Least Concern | Population trend: | Stable |
|----------------|---------------|-------------------|--------|

The Black-capped Chickadee remains abundant across its vast range, with an estimated population exceeding 100 million individuals. It adapts well to suburban and urban environments and benefits from nest box programs and winter feeding.

FEEDER TIPS

- Tube feeders or small hopper feeders work best
- Black-oil sunflower seeds are strongly preferred over striped
- Chickadees can be hand-tamed with patience over several weeks
- Place a nest box (1 1/8" hole diameter) near woodland edge
- Suet feeders are especially valued from October through March

COLORING GUIDE

| ZONE          | RECOMMENDED COLOR  |
|---------------|--------------------|
| Crown & bib   | Deep black         |
| Cheek patches | Pure white         |
| Back & wings  | Slate gray         |
| Belly         | White to pale gray |
| Flanks        | Buffy-white        |
| Bill          | Short, black       |

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Black-capped Chickadee • *Poecile atricapillus* • Module 02



## MODULE 03 / 40

# Blue Jay

*Cyanocitta cristata***QUICK STATS**

|          |                                 |
|----------|---------------------------------|
| Length   | 9.8 – 11.8 in (25 – 30 cm)      |
| Wingspan | 13.4 – 16.9 in (34 – 43 cm)     |
| Weight   | 2.5 – 3.5 oz (70 – 100 g)       |
| Lifespan | Up to 26 years (avg. 7 in wild) |

**PRESENTATION**

The Blue Jay is one of the most striking and intelligent birds to visit North American gardens. Its bold blue, white, and black plumage makes it instantly recognizable at any feeder. Despite its reputation as a bully — it dominates most other species at feeding stations — the Blue Jay plays a vital ecological role as one of the continent's most important seed dispersers. Highly adaptable and extraordinarily vocal, it produces a wide variety of calls including convincing imitations of hawks, which it uses to clear feeders of competing birds.

**HABITAT & RANGE**

The Blue Jay is found throughout the eastern half of North America, from southern Canada (Alberta east to Nova Scotia and Newfoundland) south through Florida and west to the edge of the Great Plains. It inhabits a wide range of wooded environments including deciduous and mixed forests, forest edges, suburban neighborhoods, city parks, and gardens with mature oak trees. Northern populations are partially migratory, with large flocks moving south in autumn along coastlines and ridges, though many individuals remain year-round even in cold climates.

**DIET & FEEDING**

- Acorns — by far the most important food source
- Whole and shelled peanuts — a feeder favorite
- Sunflower seeds (black-oil and striped)
- Corn (cracked or whole kernel)
- Insects, grasshoppers, beetles during summer
- Occasionally: eggs and nestlings of other birds

BEHAVIOR & SOCIAL LIFE

Blue Jays are highly social outside the breeding season, gathering in loose flocks and traveling together through the forest canopy. They are among the most vocal birds in North America, producing a rich repertoire of calls including their signature loud "jay-jay-jay" alarm, musical liquid gurgles, whisper songs, and remarkably accurate mimicry of Red-shouldered and Red-tailed Hawks. This mimicry serves multiple purposes: alerting other jays to hawk presence and, controversially, potentially clearing feeders so the jay can feed undisturbed. Blue Jays have demonstrated problem-solving abilities and tool use in laboratory settings, ranking among the most cognitively advanced birds studied.

NESTING & BREEDING

Blue Jays form monogamous pairs that often mate for life. Both sexes participate in nest building, constructing a bulky open cup of twigs, bark strips, mud, and grass in the fork of a tree or shrub, typically 10–25 feet above ground. Clutches of 2–7 eggs (usually 4–5) are incubated for 17–18 days by both parents. Nestlings are fed by both parents and fledged at 17–21 days. Pairs typically raise one brood per year. Both parents aggressively mob and chase away predators, including owls, hawks, cats, and humans.

SONG & COMMUNICATION

The Blue Jay's vocal repertoire is one of the most varied of any North American bird. Its primary call is a loud, harsh "jay" or "jeer," used as an alarm. It also produces a bell-like "tool-tool" call, soft musical whisper songs audible only at close range, and uncannily accurate imitations of at least three hawk species. Researchers believe jays use hawk mimicry both as a warning system and as a competitive strategy at feeders.

CONSERVATION STATUS

|                |               |                   |        |
|----------------|---------------|-------------------|--------|
| IUCN Red List: | Least Concern | Population trend: | Stable |
|----------------|---------------|-------------------|--------|

Blue Jay populations are stable across their range, with an estimated 45 million individuals in North America. The species has benefited from suburban expansion and the planting of oak trees in urban environments.

FEEDER TIPS

- Platform feeders or large hopper feeders accommodate their size
- Whole peanuts in the shell are irresistible and provide great viewing
- Scatter cracked corn on the ground — jays readily feed at ground level
- Plant oak trees: a single oak can produce thousands of acorns annually
- Expect jays to cache food — they may visit and depart rapidly with full beaks

COLORING GUIDE

| ZONE                | RECOMMENDED COLOR                   |
|---------------------|-------------------------------------|
| Crest & upper wings | Bright blue with black bars         |
| Tail                | Blue with black barring, white tips |
| Face & nape         | White with bold black necklace      |
| Breast & belly      | White to pale gray                  |
| Back                | Blue-gray                           |
| Bill & legs         | Black                               |

# COLOR ME!

Blue Jay • *Cyanocitta cristata* • Module 03

